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Evolution of electron localization with Co^{2+} variations in $Co_xFe_{3-x}O_4$ hollow spheres for enhanced wave absorption

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Abstract

Substitution can cause changes in electromagnetic parameters thereby affecting the electromagnetic wave (EMW) absorption performance of the material. We inferred that the fundamental reason for the performance improvement is the transitions in the absorption mechanism, which is caused by the change in electronic structure that originated from the substitution. To confirm this, we prepared $Co_xFe_{3-x}O_4$ ($0 \leq x \leq 1$) hollow microspheres and investigated its absorption performance in 2-18 GHz. Subsequently, a series of theoretical studies were carried out that focusing on the impact of increasing Co^{2+} substitution on the electrical and magnetic loss mechanisms, as well as the inherent relationship between the substitution and the absorption performance. The result indicates that in terms of dielectric loss, an increase in Co^{2+} substitution can induce a transition from conductivity loss to polarization loss by increasing the number of holes and strengthening the electron localization. Meanwhile, substitution results in a transformation from the eddy current loss to a magnetic resonance mode, and leads to an increase in magnetocrystalline anisotropy simultaneously, enhancing the magnetic loss ability. As a result, the transformation of electrical and magnetic loss mechanisms ultimately leads to an overall improvement in EMW absorption performance of $Co_xFe_{3-x}O_4$, and $CoFe_2O_4$ hollow microsphere consequently realizes an extremely high

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attenuation performance of -72.3 dB at 2.67 mm. This work provides a deeper understanding of the intrinsic relationship between the macroscopic absorption performance of ferrite and its microscopic electronic structure, which can guide the synthesis of efficient ferrite absorbers.

Keywords： Electron localization, Magnetocrystalline anisotropy, electromagnetic wave absorption, Ferrite.

1 Introduction

With the rapid development of electronic information and communication technology, multifarious intelligent wireless electronic equipment is being used, which not only facilitates people's lives but also bring non-ignorable electromagnetic pollutions [1,2]. The importance of high-efficiency absorbing materials with thin thickness, light weight, wide effective absorption band (EAB), and strong reflection loss (RL) is becoming increasingly significant [3-5]. Fe_3O_4, as a traditional magnetic absorbing material, has inevitable defects such as low magnetocrystalline anisotropy, high electrical conductivity, high density, and easy agglomeration. Structural regulation, such as flower-, urchin-, rugby-like [6-8], and foam, hollow sphere structure [9,10] can reduce its density and agglomeration. Meanwhile, doping or substitution of Fe atoms with Gd [11], Zr [12], Cu [13,14], Mn [15,16] can improve the EAB and RL values furtherly, thereby improving the EMW absorption performance. As a result, a large amount of experimental research on $M_xFe_{3-x}O_4$ can be found in the literature [17,18], among which $Co_xFe_{3-x}O_4$ is widely investigated due to its high stability, corrosion resistance, and excellent magnetic properties [19-21].

The replacement content, i.e. the value of x, is crucial for the absorption performance of $M_xFe_{3-x}O_4$ [22,23]. Since the evaluation of material absorption performance is mostly calculated from electromagnetic parameters based on transmission line theory. The value of x will inevitably affect the electromagnetic parameters of the material, thereby affecting its absorption performance [24].

Theoretically, the change in electromagnetic parameters caused by substitution should originate from the change in the electronic structural properties. The dielectric loss mechanism, such as conductivity loss, dipole polarization loss, interface polarization loss and the magnetic loss mechanism, which including eddy current loss, natural resonance, exchange resonance are closely related to the electronic structural properties [25,26]. In other words, the intrinsic reason for the improvement in absorption performance caused by substitution is the change in electromagnetic loss mechanism. There are reports on substitution that change the loss mechanism. For example, Zhang et al. revealed that the use of N substituted doping in graphite will lead to the breaking of its structural symmetry and the separation of space charge, thus achieving a shift in the conductivity loss to a stronger dipole polarization loss [27]. Ren et al. established a bi-directional etching substitution strategy to modulate the chemical composition and heterostructure of Zn/Co hybrids, and the results showed that substitution etching of Zn on Co-ZIF was able to achieve the transition from eddy current loss to magnetic resonance [28]. Li et al. used Cu and Ni for substitutional doping of Co_3O_4, respectively, revealing that the addition of Cu can induce the generation of polarization centers and thus increase the defect polarization, while the addition of Ni can bring about a certain degree of eddy current loss [29]. However, due to limitations in its own properties and the lack of advanced testing equipment, it is still difficult to reveal the underlying reasons between ferrite substitution and loss mechanisms experimentally. DFT theoretical calculations can break through the limitations of experimental methods in predicting and analyzing micro mechanisms, and therefore are widely used in various fields [30,31]. However, few reports on mechanism analysis by using DFT can be found in the field of electromagnetic absorption.

In this work, we synthesized $Co_xFe_{3-x}O_4$ hollow microsphere by using a simple and facile hydrothermal method by adjusting the Co substitution content, and investigated its EMW absorption performance in the range of 2-18 GHz. Subsequently, in-depth exploration of its electromagnetic parameters and DFT theoretical calculations were carried out, with a focus on exploring the changes in material electronic structure

and the absorption mechanisms with Co content variations, as well as the inherent connections between them. It is regard to reveal the intrinsic relationship between the absorption performance and electronic structure of $Co_xFe_{3-x}O_4$, which has theoretical and experimental significance for the synthesis and regulation of efficient ferrite electromagnetic absorption materials.

2 Experimental procedures

2.1 Reagents

Cobalt Nitrate Hexahydrate ($Co(NO_3)_2·6H_2O$), Iron Nitrate Nine Hydrate ($Fe(NO_3)_3·9H_2O$), Urea (CH_4N_2O), Trisodium Citrate Dihydrate ($C_6H_5Na_3O_7·2H_2O$) and Polyacrylamide (PAM) were purchased from Aladdin Industrial Corporation. All chemical reagents were used as received without any further purification.

2.2 Synthesis of $Co_xFe_{3-x}O_4$ hollow spheres

$Co_xFe_{3-x}O_4$ ($0 \le x \le 1$) were prepared using a facile hydrothermal method (Fig. S1). Typically, an appropriate amount of $Fe(NO_3)_3·9H_2O$, 1.76 g $C_6H_5Na_3O_7·2H_2O$ and 0.54 g CH_4N_2O were dissolved in 60 ml deionized water successively under continuous stirring at room temperature until the solution turns uniformly green. Subsequently, 0, 0.25, 0.5, 0.75 and 1 mmol $Co(NO_3)_2·6H_2O$ was added to the above solution, respectively, when the total mole of Co and Fe is 3 mmol in the products. Finally, after adding 0.45 g PAM and under 40 min stirring, the above solution was transferred to a 100 ml Teflon lined stainless-steel autoclave and treated at 200℃ for 12 h. After cooling to room temperature, the products were collected by centrifugation and washed with ethanol several times before drying at 80 ℃ for 12 h. The dried product was collected and ground into a black powder. According to the set molar ratio, the products were named as Fe_3O_4 (Co:Fe=0:1), $Co_{0.25}Fe_{2.75}O_4$ (Co:Fe=1:11), $Co_{0.5}Fe_{2.5}O_4$ (Co:Fe=1:5), $Co_{0.75}Fe_{2.25}O_4$ (Co:Fe=1:3) and $CoFe_2O_4$ (Co: Fe=1:2).

2.3 Characterization

The morphology of the samples was investigated using a field emission scanning electron microscopy (SEM; Quanta × 50 FEG) and a transmission electron microscopy (JEM-2100F TEM operating at accelerating voltage of 200 kV was used for TEM,

HRTEM). The element dispersion was analyzed using X-ray energy dispersive spectroscopy (EDS) with an accelerating voltage of 15 kV. The crystalline structure was characterized by X-ray diffraction (XRD, Bruker D8 ADVANCE instrument with a Cu/K-alpha radiation from 10° to 80° (λ = 0.15418 nm)). The element valence analysis was carried out through X-ray photoelectron spectroscopy (XPS, Thermo SCIENTIFIC K-ALPHA with Al-Kα radiation). The magnetic properties were measured at room temperature on a vibrating sample magnetometer (VSM, Lake Shore 7404) with a magnetic field up to 2.0 T.

2.4 Electromagnetic measurements

The electromagnetic parameters over 2-18 GHz were measured on a network vector analyzer (Anritsu MS46322B, Japan) based on the coaxial method. All the products were uniformly mixed with paraffin (50 wt.%) and suppressed into a toroid (Φ_{out} = 7 mm, Φ_{in} = 3.04 mm) with a thickness of 3.0 mm. Then, according to the transmission line theory, RL values were calculated by the following formula [32]:

$$Z_{in} = Z_0\sqrt{\frac{\mu_r}{\varepsilon_r}}\tanh\left(j\frac{2\pi fd}{c}\sqrt{\mu_r\varepsilon_r}\right) \quad (1)$$

$$RL(dB) = 20lg\left|\frac{Z_{in}-Z_0}{Z_{in}+Z_0}\right| \quad (2)$$

where Z_{in} is the input impedance, Z_0 is the impedance of free space, d, f and c are the thickness of the film filled with absorber, the frequency, and the speed of light in vacuum.

2.5 Calculation method

The Density Functional Theory (DFT) calculations were performed using the Vienna ab-initio Simulation Package (VASP). The exchange-correlation functional were describe using Perdew-Burke-Ernzerhof (PBE) within the Generalized Gradient Approximation (GGA). The PAW potential function was used to describe the electron-ion interactions. The 5 × 5 × 5 and 11 × 11 × 11 k-point grid were applied to the Brillouin zone for structural optimization and electronic structure calculations, respectively. The energy cutoff was conducted as 500 eV, the maximum force and energy tolerance accuracy were set as 0.02 eV/Å, 1×10^{-6} eV/atom, respectively. Herein, a 2 × 1 × 1 surpercell (28 atoms) of $Co_xFe_{3-x}O_4$ were constructed for study. In order to avoid

ignoring the strong correlation of 3d electrons, the DFT+U correction method was introduced, in which the U values of Co and Fe were set to 4.22 and 4.08, and the J values were set to 0.8 and 0.79 [33].

3 Results and discussion

3.1 Morphological and structural characterizations

Figure 1 shows the XRD patterns of all the five products. As can be seen from Fig. 1(a), five characteristic peaks at 2Theta = 30.1°, 35.4°, 43.1°, 56.9° and 62.6° are assigned to the (220), (311), (400), (511), and (440) planes of cubic inverse spinel phase $Co_xFe_{3-x}O_4$ (PDF # 22-1086) [34, 35]. No characteristic peaks for other impurities were obtained, implying the products have high purity. As is shown, Co^{2+} doping induced a slight shift of the diffraction peaks towards higher angles, the reason for this is that the Co^{2+} radius (0.745 Å) is smaller than that of Fe^{2+} (0.78 Å), resulting in a decrease in interplanar spacing. It is worth noting that with the increase of Co content, the average grain size calculated by Scherrer's formula decreases slightly (Fig. S2), indicating there is a difference in crystallinity under the same preparation process with the variation of Co^{2+} content.

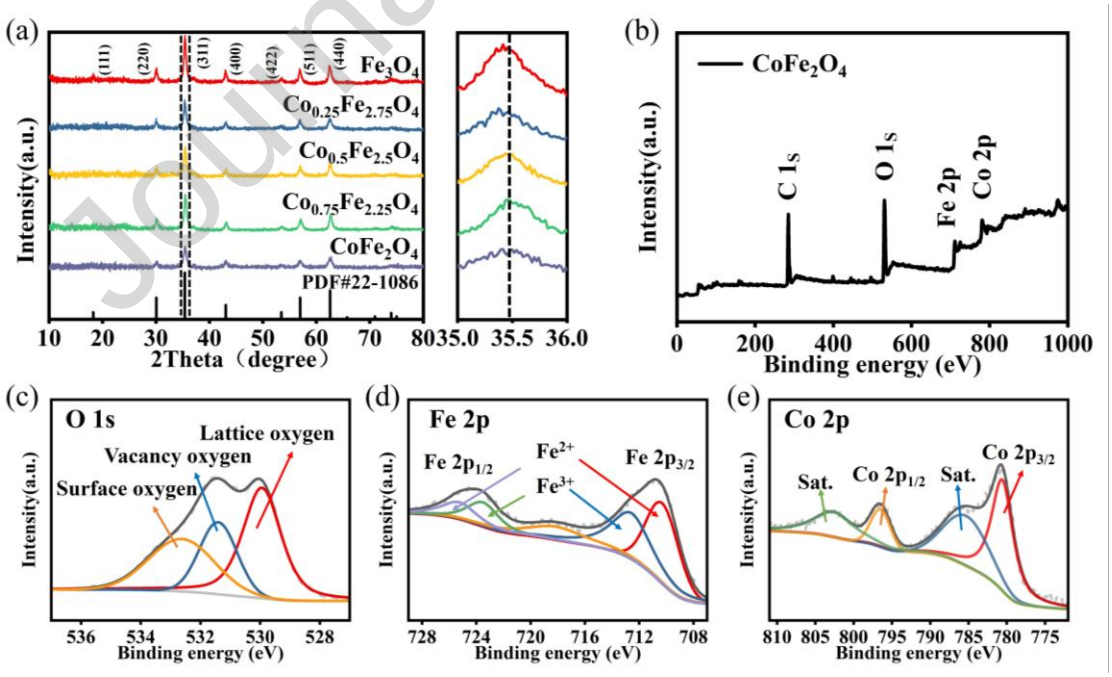


Figure 1 (a) XRD patterns of the $Co_xFe_{3-x}O_4$. (b) Full scan XPS spectrum, (c) O 1s spectrum, (d) Fe 2p spectrum of $CoFe_2O_4$, (e) Co 2p spectrum.

In order to investigate the chemical bonding and chemical composition of the $Co_xFe_{3-x}O_4$, $CoFe_2O_4$ sample was further analyzed using XPS. Figure 1(b) suggests that Co, Fe, and O were present in $CoFe_2O_4$ sample, which is consistent with the results of XRD results. The high elemental oxygen content is evidenced by a distinct O1s peaks in the full spectra. In Figure 1(c), the high resolution spectrum of O1s could be divided into lattice oxygen (529.9 eV),vacancy oxygen (531.4 eV) and surface oxygen (532.7 eV). For the Fe2p spectra (Fig. 1(d)), the two main peaks at 710.8 and 724.2 eV correspond to $Fe2p_{3/2}$ and $Fe2p_{1/2}$, respectively, and the fitted peaks reveal the coexistence of Fe^{3+} and Fe^{2+} [36]. Similarly, the peaks at 780.6 eV and 785.3 eV respond to Co $2p_{3/2}$ and Co $2p_{1/2}$, respectively, which are typical spin orbitals for Co element. Furthermore, the peaks at 796.5 eV and 802.6 eV originate from two satellite peak oscillation effects (Fig. 1(e)) [37].

Figures 2(a)-2(f) show the SEM images of the prepared samples. In general, all the samples possess a regular spherical microstructure which is uniform and monodisperse. However, with the variation of Co^{2+} content, the volume, size distribution, and microstructure of the nanospheres exhibit differences. As is shown Fig. 2(f) and Fig. S3, with the increasing of *x* from 0 to 1, the average size for the samples is 200 nm, 250 nm, 180 nm, 240 nm and 230 nm, respectively. The surfaces of microspheres with x = 0, 0.25, and 1 are smooth, while those with x = 0.5 and 0.75 are relatively rough. This difference is associated with the crystal coordination environment [38]. This is consistent with the XRD results, confirming that variations of Co^{2+} can affect the crystallinity of the product. It is worth noting that compared with other samples, the size distribution of $Co_{0.75}Fe_{2.25}O_4$ is significantly uneven, which is considered be beneficial for the EMW absorption by generating multiple interfaces [39]. The EDS (Fig. S4) results confirmed that as the value of *x* gradually increased from 0 to 1, the detected surface component of the as-prepared microspheres is $Fe_{2.861}O_4$, $Co_{0.243}Fe_{2.757}O_4$, $Co_{0.567}Fe_{2.433}O_4$, $Co_{0.732}Fe_{2.268}O_4$ and $Co_{0.918}Fe_{2.082}O_4$, respectively (Table S1). This, together with XRD, confirms the successful preparation of preset

$Co_xFe_{3-x}O_4$. In addition, by comparing SEM images, it can be observed that $CoFe_2O_4$ exhibits uniform size distribution and excellent surface smoothness among the samples.

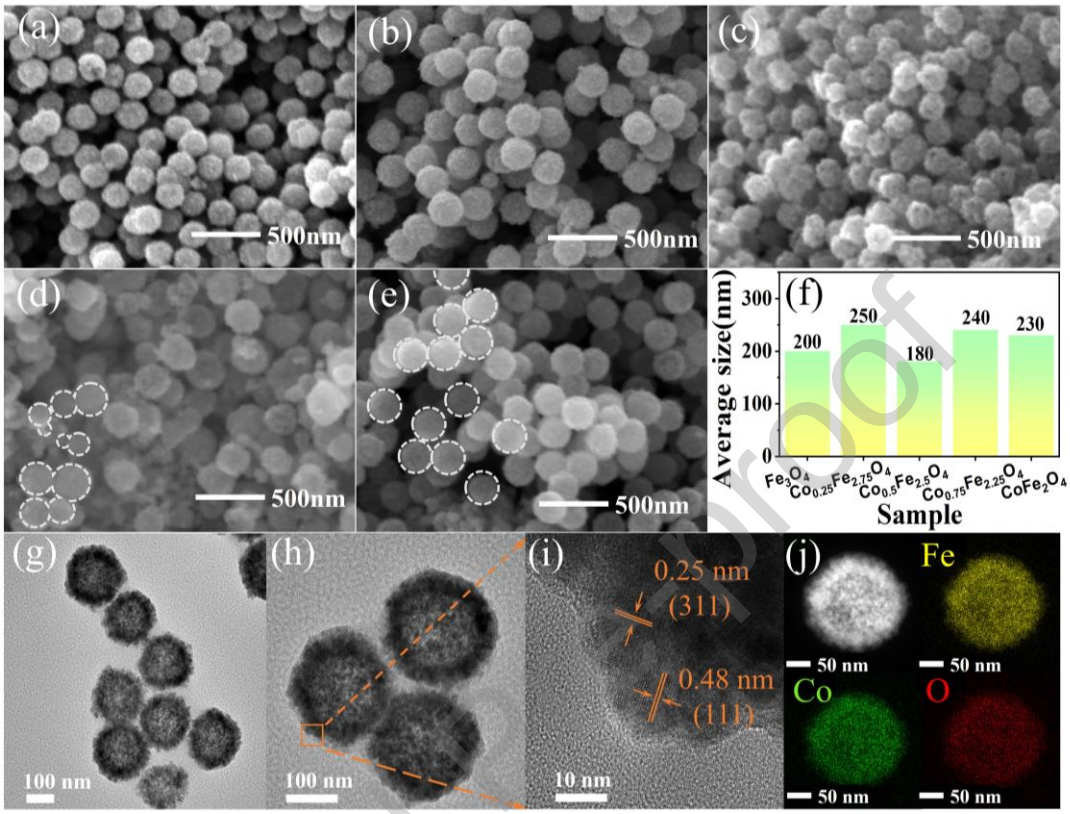


Figure 2 SEM images of the $Co_xFe_{3-x}O_4$ hollow nanospheres: (a) x = 0, (b) x = 0.25, (c) x = 0.5, (d) x = 0.75, (e) x = 1. (f) The average size of $Co_xFe_{3-x}O_4$ hollow nanospheres. (g, h) TEM, (i) HRTEM and (j) EDX image of $CoFe_2O_4$ hollow nanospheres.

TEM images further confirm that the prepared $Co_xFe_{3-x}O_4$ possess a perfect hollow microsphere structure, which can reduce density and improve the absorption efficiency by creating multiple reflection and scattering propagation paths for incident EMW simultaneously. As is shown in Figs. 2(g) and 2(h), the size distribution of $CoFe_2O_4$ microspheres is uniform, with an overall size of about 230 nm, which is consistent with the SEM results. The obvious color difference confirms the sphere has a stable hollow structure, with a uniform shell thickness of about 24 nm. The interplanar spacings have a spacing of 0.25 nm and 0.48 nm, corresponding to the (311) and (111) planes of $CoFe_2O_4$, respectively (Fig. 2(i)). EDX mapping results indicate that Co, Fe and O are uniformly distributed (Fig. 2(j)), reaffirming the uniformity of the microsphere shell.

More TEM information about $Co_xFe_{3-x}O_4$ can be found in Fig. S5.

3.2 EMW absorption performance of $Co_xFe_{3-x}O_4$ composites

The EMW absorption property of all the samples were investigated. Figures 3(a)-3(e) show the RL values and EAB for each sample within a thickness range of 1-3.5 mm. The optimal EAB, RL values and their corresponding thicknesses for each sample are shown in Fig. 3(f). The 3D performance images are shown in Fig. S6. The single Fe_3O_4 exhibits a minimum RL (RL_{min}) value of -18.2 dB at 1.5 mm thickness, which decreases to -13 dB at a thickness of 2 mm for the $Co_{0.25}Fe_{2.75}O_4$ (Figs. 3(a) and 3(b)). However, $Co_{0.5}Fe_{2.5}O_4$ and $Co_{0.75}Fe_{2.25}O_4$ deliver an enhanced RL_{min} value of -15.7/-46.1 dB at 2.5/3.1 mm thickness, respectively (Figs. 3(c) and 3(d)). Notably, the $CoFe_2O_4$ exhibits the optimal absorption performance among the five samples, achieving a RL_{min} value of −72.3 dB at 2.67 mm thickness (Fig. 3(e)). The EAB values of $Co_xFe_{3-x}O_4$ are 2.08 GHz (x = 0), 3.2 GHz (x = 0.25), 2.48 GHz (x = 0.5), 3.2 GHz (x = 0.75) and 3.2 GHz (x = 1), respectively (Fig. 3(f)). As x gradually changes from 0 to 1, the absorption performance decreased initially and increased afterward. In order to find out the intrinsic relationship between changes in absorption performance and Co content, we conducted in-depth analysis of electromagnetic parameters.

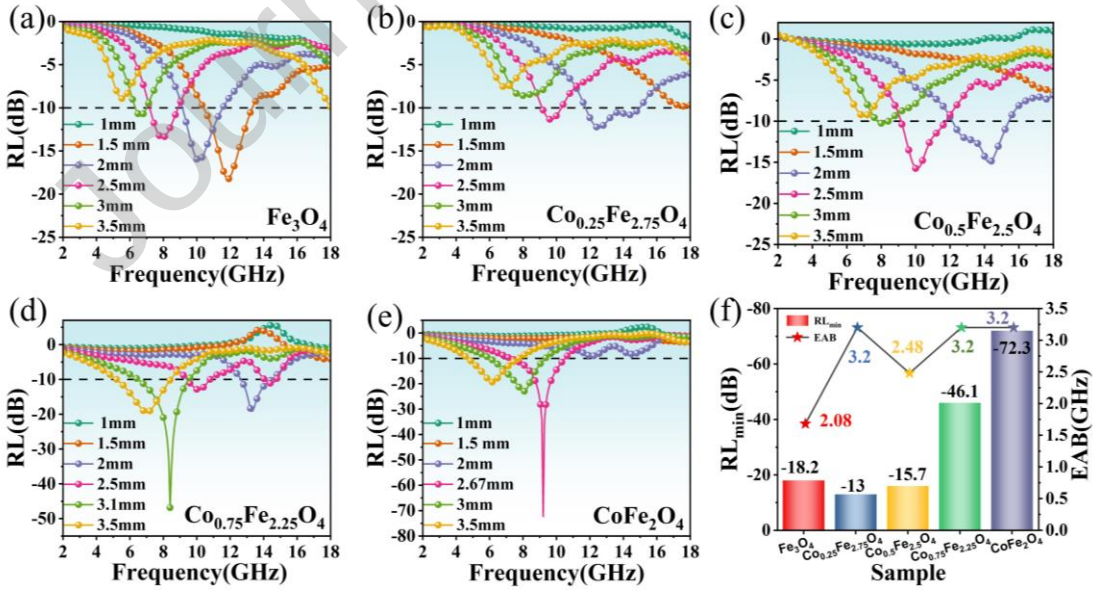


Figure 3 RL values of (a) Fe_3O_4, (b) $Co_{0.25}Fe_{2.75}O_4$, (c) $Co_{0.5}Fe_{2.5}O_4$, (d) $Co_{0.75}Fe_{2.25}O_4$ and (e) $CoFe_2O_4$ with the thickness from 1 mm to 3.5 mm; (f) RL_{min} and EBA of $Co_xFe_{3-x}O_4$.

3.3 EMW absorption mechanisms of $Co_xFe_{3-x}O_4$ composites

Figures 4(a)-4(c) show the complex permittivity ($\varepsilon_r = \varepsilon' - j\varepsilon''$) and the dielectric loss tangent ($tan\delta_\varepsilon$) of the five samples. As is shown, for Fe_3O_4, $Co_{0.25}Fe_{2.75}O_4$ and $Co_{0.5}Fe_{2.5}O_4$, the value of ε' is relatively constant in the frequency range of 2-11 GHz, and then decreases with the increasing frequency. The ε' of $Co_{0.75}Fe_{2.25}O_4$ and $CoFe_2O_4$ possesses more pronounced dispersion behavior. Between 13 GHz and 16 GHz, the values decrease rapidly from 13.1 and 11.9 to 3.5 and 6.9, respectively. Relaxation peaks can be observed in Fig. 4(b) for all samples, indicating the exhibition of Debye dielectric relaxation process, and the relaxation behavior becomes more pronounced with the increase of Co content. Figrue 4(c) represents the values of $tan\delta_\varepsilon$ versus *f*, where all samples show higher $tan\delta_\varepsilon$ values in the frequency range of 12-18 GHz, indicating that dielectric loss plays a major role in mid and high frequencies. It is worth noting that $tan\delta_\varepsilon$ of $Co_{0.75}Fe_{2.25}O_4$ is significantly higher than that of the other samples at around 14 GHz. This can be attributed to the size heterogeneity of $Co_{0.75}Fe_{2.25}O_4$, as confirmed by SEM, which can promote charge aggregation at the interface and lead to stronger interfacial relaxation polarization. Figure 4(d) represents the average permittivity values of all the samples. It is interesting that with *x* gradually increases from 0 to 0.5, the complex permittivity decreased initially and increased afterward, which is in consistent with the absorption performance of the samples.

Based on the outcomes provided, it can be deduced that the apparent fluctuation in the complex permittivity is significantly associated with the Co^{2+} incorporation in the $Co_xFe_{3-x}O_4$ absorbers. In order to examine the dielectric loss mechanisms in the samples, Cole-Cole plots are typically utilized for analysis [40]. According to the Debye theory, the relationship between ε' and ε'' can be described by the following equation [41]:

$$(\varepsilon' - \frac{\varepsilon_s+\varepsilon_\infty}{2})^2 + (\varepsilon'')^2 = (\frac{\varepsilon_s-\varepsilon_\infty}{2})^2 \quad (3)$$

where ε_∞ is the high-frequency permittivity, ε_s is static permittivity. It is obvious that for Fe_3O_4 and $Co_{0.25}Fe_{2.75}O_4$, the Cole-Cole image contains several irregular curves, and the curve shapes are almost identical (Figs. 4(e) and 4(f)). According to the Debye relaxation theory [42]. The dielectric loss mechanism of both is dominated by

conductivity loss. Conductivity loss still exists in $Co_{0.5}Fe_{2.5}O_4$, but its Cole-Cole image shows a clear semi-circular curve, which indicates the occurrence of a polarization loss mechanism (Fig. 4(g)) [43]. When *x* reaches 0.75, the conductivity loss disappears in $Co_{0.75}Fe_{2.25}O_4$ and there are only three distinct semicircles in the Cole-Cole curve (Fig. 4(h)). For $CoFe_2O_4$, the number of semicircles increases, implying the existence of a multiple dielectric polarization mechanisms (Fig. 4(i)) [44].

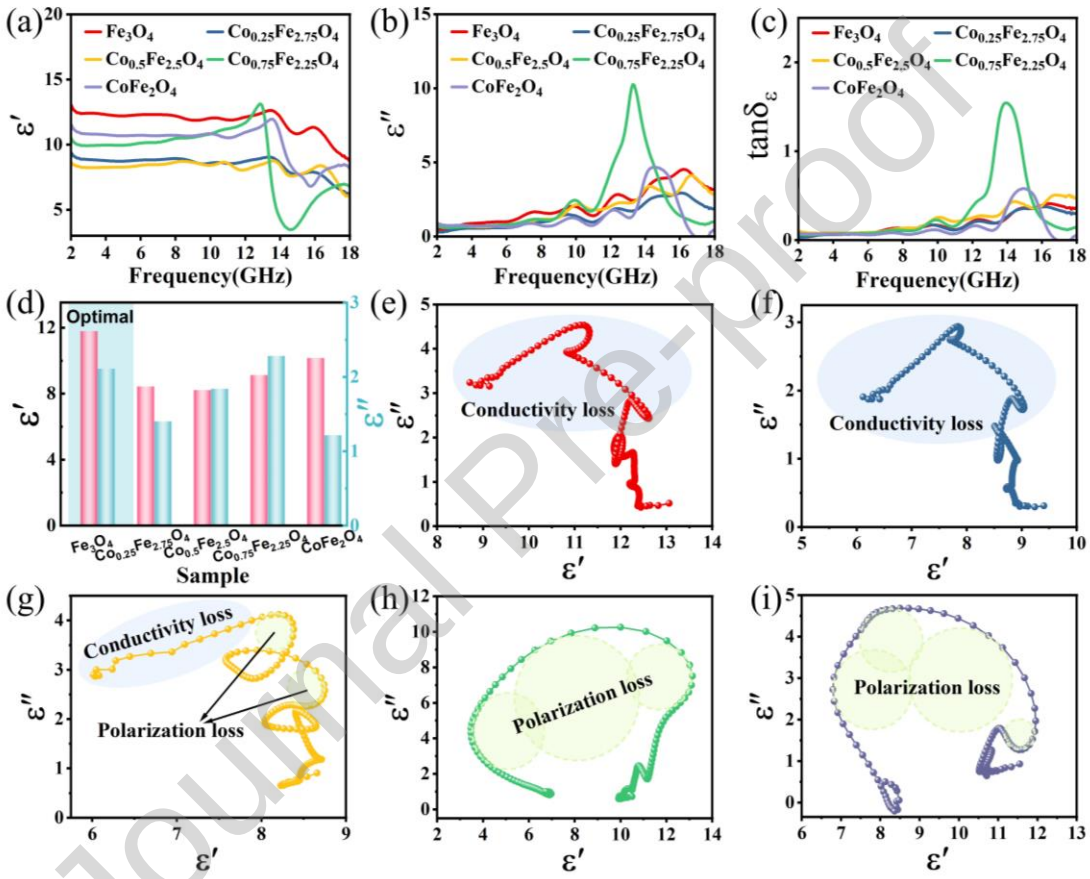


Figure 4 (a) The real and (b) imaginary part of relative complex permittivity, (c) dielectric loss tangent of five samples, (d) the average values for ε' and ε'' of five samples, the Cole-Cole plots of (e) Fe_3O_4, (f) $Co_{0.25}Fe_{2.75}O_4$, (g) $Co_{0.5}Fe_{2.5}O_4$, (h) $Co_{0.75}Fe_{2.25}O_4$ and (i) $CoFe_2O_4$.

The above results indicate that in hollow sphere $Co_xFe_{3-x}O_4$, with the Co^{2+} variations, the dielectric loss mechanism undergoes a transition from conductivity loss dominate to conductivity-polarization relaxation loss co-dominate and finally polarization relaxation loss dominate process. To gain a deeper understanding of transformation mechanism, the electronic structures of $Co_xFe_{3-x}O_4$ were detected by the DFT calculations. The density of states (DOS) of $Co_xFe_{3-x}O_4$ are shown in Figs. 5(a)-

5(c) and Fig. S7. As is shown, with the Co content increases, the total DOS of $Co_xFe_{3-x}O_4$ at the Fermi level decreases until it vanishes at x = 1, and the system becomes insulating with a band gap of 0.42 eV. The electron localization function (ELF) is used to quantitatively describe the degree of electron localization and the type of bonding between atoms. Figures 5(d)-5(f) and Fig. S8 show ELF of (100) crystal face for the five samples (from blue to red indicates a gradual strengthening of electron localization). The electrons of all samples are located around the O atoms, suggesting that ionic bonds are formed between O-Fe and O-Co, which is consistent that $Co_xFe_{3-x}O_4$ is an ionic compound. In addition, as the Co content increases, the area of the electron cloud surrounding the octahedral metal cation sites decreases, meanwhile the ELF value at the central position decreases from 10.2E-3 to 7.01E-3, suggesting that Co doping strengthen the degree of electron localization furtherly.

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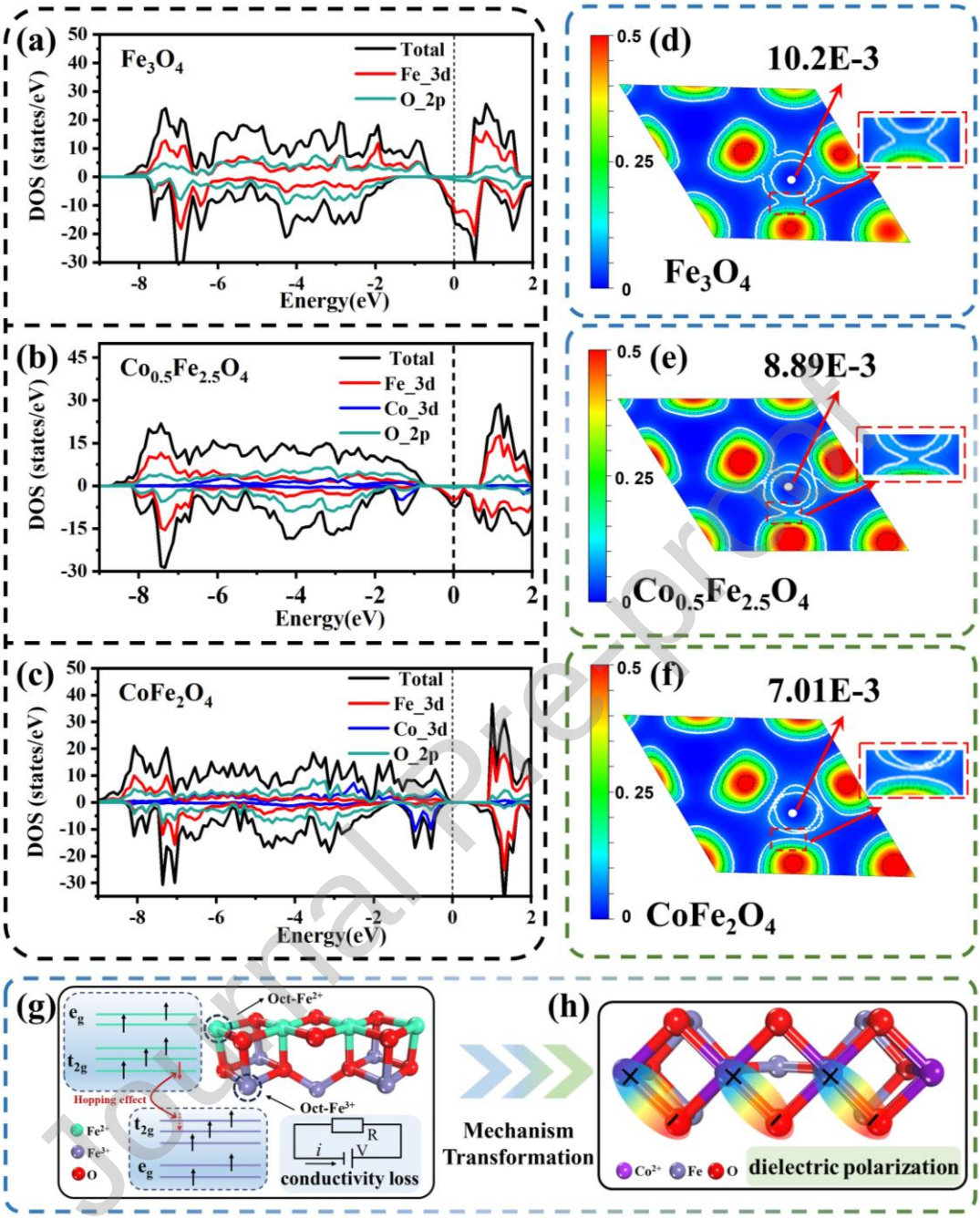


Figure 5 Density of states (DOS) of (a) Fe_3O_4, (b) $Co_{0.5}Fe_{2.5}O_4$, (c) $CoFe_2O_4$ (The Fermi level is set to zero energy and indicated by the vertical dotted line). Electron localization function (ELF) crystal surface (100) of (d) Fe_3O_4, (e) $Co_{0.5}Fe_{2.5}O_4$, (f) $CoFe_2O_4$. Possible dielectric loss mechanisms (g) conductivity loss, (h) dielectric polarization in $Co_xFe_{3-x}O_4$ samples.

Based on the provided information, it appears that when $x = 0$, Fe_3O_4 possesses the ability to generate a significant number of free electrons through the "hopping effect" between Fe^{2+} and Fe^{3+} ions at the octahedral position (Fig. 5(g)), leading to the possession of half-metallic properties [45, 46]. When an external electromagnetic field

is induced, these free electrons can form an electric current and consume the energy of the incident EMW in the form of Joule heat, resulting in conductivity loss. However, when x is equal to 0.25, the substitution of Fe^{2+} by Co^{2+} would weaken the "hopping effect", which results in the decrease of the conductivity. According to free electron theory: $\varepsilon'' = \sigma/2\pi f\varepsilon_0$, where ε_0 is the permittivity of free space, f is the EMW frequency, and σ is the conductivity [47, 48]. This results in a reduction of the complex permittivity of $Co_{0.25}Fe_{2.75}O_4$ in comparison to Fe_3O_4. But when x reaches 0.5, the number of holes generated by Co^{2+} substitution increases [49]. In addition, the ELF decreased by 12.8% which means the electron localization is reinforced furtherly. Therefore, the number of dipoles formed by electrons and holes increases. At the same time, the enhancement of localization can significantly widen the spacing between electrons and holes, which induces an enlargement in the electric dipole moment of the intrinsic electric dipole according to definition $p = q \times l$ (p is the electric dipole moment, q is the charge, l is the spacing). This will lead to an increase in dipole polarization, the sample exhibits a dielectric loss mechanism characterized by the coexistence of conductivity loss and dipole polarization loss. Consequently, the corresponding complex permittivity increases [50]. The discussion above is agreed with the dielectric properties of samples shown in Fig. 4(d). When x = 0.75, due to the reduction of Fe^{2+}, the "hopping effect" is weakened. Therefore, the conductivity loss decreases while the dipole polarization increases continuously. When x = 1, the electron localization is the highest, and $CoFe_2O_4$ transforms into a more complex polarization relaxation mechanism (Fig. 5(h)).

Figures 6(a) and 6(b) show the complex permeability ($\mu_r = \mu' - j\mu''$) of five the samples. As can be seen, the frequency response characteristics of μ' with higher Co^{2+} are more pronounced, which is consistent with the performance of the complex permittivity. The imaginary part μ'' of the five samples, which represents the magnetic loss ability, shows significant differences. For example, the μ'' values of x = 0 and x = 0.25 are almost consistent within 2-18 GHz. Inconsistency in performance was found at low and high frequency for x = 0.5. Based on SEM and other factors, this difference may be related to increased Co. The starting μ'' values of x = 0.75 and x = 1 are significantly higher than that of the other three samples, showing a downward trend

with the increasing frequency and exhibiting negative values at 11.04-16.72 GHz and 13.3-16.3 GHz, respectively. According to Lenz law, the negative value of μ'' is probably originated from the induced electromagnetic field which can counteract the incident EMW, and consume its energy [51, 52]. In addition, at the corresponding frequency, the ε'' of the $Co_{0.75}Fe_{2.25}O_4$ and $CoFe_2O_4$ reaches a maximum indicating the presence of significant electromagnetic coupling within the absorber (Fig. 4(b)), which significantly promotes the conversion between electromagnetic energies [53, 54]. Therefore, it can be inferred that the samples with $x = 0.75$ and $x = 1$ have an enhanced induction ability towards incident EMW, and correspondingly possess an enhanced magnetic loss ability. In addition, resonance peaks were observed on the μ'' curves of all samples. The resonance peaks for Fe_3O_4 and $Co_{0.25}Fe_{2.75}O_4$ are located at ~11.2 GHz, that for $Co_{0.5}Fe_{2.5}O_4$ is located at 15.52 GHz, while those for $Co_{0.75}Fe_{2.25}O_4$ and $CoFe_2O_4$ are located at ~17.28 GHz. The above values indicate that as the Co content increases, the hollow microspheres $Co_xFe_{3-x}O_4$ possess a decreasing magnetic storage capacity and an increasing magnetic consumption capacity, which is also confirmed by $tan\delta_\mu$ (Fig. 6(c)).

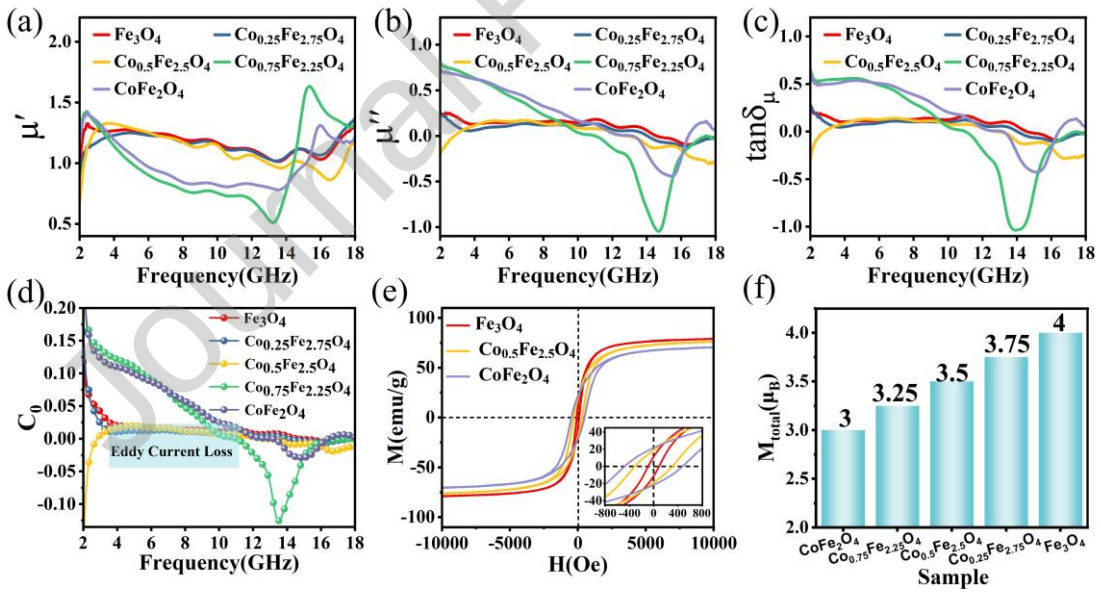


Figure 6 (a) The real and (b) imaginary part of relative complex permeability, (c) magnetic loss tangent, (d) eddy current loss of $Co_xFe_{3-x}O_4$, (e) magnetic hysteresis loops of $Co_xFe_{3-x}O_4$ at room temperature, (f) the total magnetic moment of each molecular formula of $Co_xFe_{3-x}O_4$.

To understand the mechanism of magnetic loss and its variation with Co content, we conducted a study on its C_0 value:

$$C_0 = \mu''\mu'^{-2}f^{-1} = \frac{2}{3}\pi\mu_0 d^2\sigma \quad (4)$$

in which μ_0 and σ are vacuum permeability and conductivity, respectively. As is shown in Fig. 6(d), C_0 values remain almost unchanged within 3.84 - 9.36 GHz for x = 0, 0.25 and 0.5, indicating that eddy current loss is the main magnetic loss mechanism [55]. This is consistent with the previous inference: due to the presence of substantial free electrons in those samples, low-frequency electromagnetic fields can easily induce the generation of eddy currents and consume incident EMW in the form of heat generation. However, as the frequency increases, the magnetic loss mechanism switches to magnetic resonance mode, which mainly includes natural resonance and exchange resonance. When x reaches 0.75 and 1, the number of free electrons decreases and the degree of localization increases, which can suppress the generation of eddy currents. Therefore, the magnetic loss mode in the whole frequency range is magnetic resonance mode. According to the reports [56, 57], when the magnetic particles size is below 100 nm, the exchange energy will greatly surpass the magnetostatic energy, making the exchange resonance becomes the main form of magnetic resonance. Considering the size of the hollow microspheres in this work is all exceeds 100 nm, we inferred that the natural resonance should be the main magnetic loss mechanism in this study, which is considered originates from magnetocrystalline anisotropy.

To further investigate the natural resonance behavior of absorber. Hysteresis loops of the samples at room temperature are evaluated in Fig. 6(e). The saturation magnetizations (M_s) of the products are 80.35 emu/g, 77.8 emu/g, and 74.9 emu/g, respectively. The negative correlation between *Ms* and Co concentration is in agreement with the first-principles calculations (Fig. 6(f) and Table S2), which can be attributed to the ferrimagnetic structure of $Co_xFe_{3-x}O_4$ [58, 59]. The increased coercivity values of $Co_xFe_{3-x}O_4$ from 91.15 Oe, 311.25 Oe to 481.32 Oe, indicates a more pronounced magnetocrystalline anisotropy of $Co_xFe_{3-x}O_4$ with the increasing of Co concentration, which can be explained by the Law of approach to saturation (LAS) method [60]. According to the LAS method, the approach saturated M are calculated as follows [61]:

$$M = M_S\left[1 - \frac{b}{H^2}\right] \quad (5)$$

where M is the magnetization, H is the applied magnetic field, M_s is the saturation

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magnetization, and the slope b is given by:

$$b = \frac{8k_1^2}{105\mu_0^2 M_S^2} \quad (6)$$

in which μ_0 is vacuum permeability, k_1 is the magnetocrystalline anisotropy constant, the numerical coefficient 8/105 holds for random polycrystalline samples with cubic anisotropy. The variations of the magnetocrystalline anisotropy field (H_k) between samples were compared by calculating the value of K_1. From Eqs. (5) and (6), the values of K_1 are 4.74E+05 J/m^3, 3.18E+05 J/m^3, and 0.96E+05 J/m^3 for $CoFe_2O_4$, $Co_{0.5}Fe_{2.5}O_4$ and Fe_3O_4, respectively, confirming an increase of magnetocrystalline anisotropy at higher Co concentration [62]. According to the natural resonance model of a cubic crystal [63]:

$$f_r = \frac{\omega_r}{2\pi} = \frac{\gamma}{2\pi} H_k \quad (7)$$

in which f_r is the intrinsic resonance frequency, γ is the spin-to-magnet ratio. Therefore, the larger the H_k, the higher the natural resonance frequency. This is consistent with the phenomenon that the resonant frequency in Fig. 6(b) gradually shifts towards higher frequency with the increase of Co content, indicating a rationality of the above inference.

The above results indicate that in the hollow microspheres $Co_xFe_{3-x}O_4$ prepared in this work, the increase of Co^{2+} will cause a shift of the magnetic loss mechanism from eddy current loss to magnetic resonance mode. The magnetocrystalline anisotropy will be enhanced simultaneously, converting more EMW energy into magnetization energy, facilitating the dissipation of incident EMW [64].

In order to further insight into the cause of the excellent EMW attenuation behavior of $Co_xFe_{3-x}O_4$ samples, the impedance matching properties of the absorber were calculated in Fig. 7(a, b) and S9(a-c). For $Co_{0.75}Fe_{2.25}O_4$ and $CoFe_2O_4$, the $|Z_{in}/Z_0|$ values in the 8.0 - 9.2 GHz and 8.64 - 11.04 GHz frequency ranges are close to 1, but the RL_{min} only occur at 8.4 GHz and 9.2 GHz, respectively, which can be explained through the phase cancellation model in the wave cancellation theory (Fig. S10(a) and S10(b)) [65, 66]. According to Eqs. S(1) and S(3) (supplementary material), both $R_M(d^-)$ and $V(t,d^-)/V(i,d^-)$ of $Co_xFe_{3-x}O_4$ have been calculated and shown in Figures 7(c) and

7(d) and Figures S9(d) - S9(f). RL_{min} for all samples occurred when the phase difference between beams *t* and *r* was approximately π. Furthermore, the difference between the values of $|R_M(d^-)|$ and $|V(t,d^-)/V(i,d^-)|$ is the smallest (0.0002) for $CoFe_2O_4$ as compared to the other samples, which consequently realizes an extremely high attenuation performance of −72.3 dB (Table S3). These results are in good agreement with those of the wave cancellation theory [32, 67].

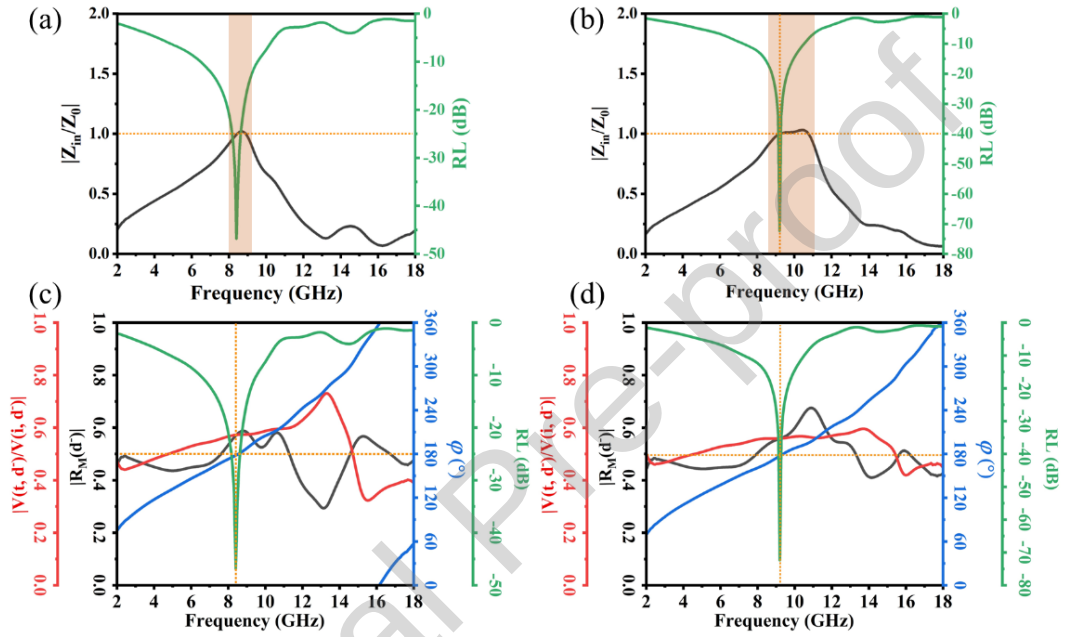


Figure 7 RL_{min}(dB) curve and $|Z_{in}/Z_0|$ values of (a) $Co_{0.75}Fe_{2.25}O_4$ and (b) $CoFe_2O_4$; RL_{min}(dB) curves, phase difference curves, $R_M(d^-)$ curves, and $|V(t,d^-)/V(i,d^-)|$ curves of (c) $Co_{0.75}Fe_{2.25}O_4$ and (d) $CoFe_2O_4$ (where $R_M(d^-)$ is the reflection coefficient, $|V(t,d^-)/V(i,d^-)|$ is the voltage ratio of the total transmitted beam to the incident beam).

Based on the above analysis, the impact of Co^{2+} variations were mainly reflected in the regulation of dielectric loss and magnetic loss (Fig. 8). In terms of dielectric loss mechanism, Fe_3O_4 and $Co_{0.25}Fe_{2.75}O_4$ present conductivity loss, $Co_{0.5}Fe_{2.5}O_4$ exhibits both conductivity loss and polarization loss, while $Co_{0.75}Fe_{2.25}O_3$ and $CoFe_2O_4$ exhibit polarization loss. Theoretical research indicates that the increasement of Co^{2+} substitution can enhance the dipole distance and lead to stronger dipole polarization by increasing the number of holes and strengthening the electron localization, thus inducing the transition of electrical loss mechanism from conductivity loss to polarization loss. In terms of magnetic loss, the eddy current loss is significant in Fe_3O_4,

$Co_{0.25}Fe_{2.75}O_4$, and $Co_{0.5}Fe_{2.5}O_4$, while the performance of $Co_{0.75}Fe_{2.25}O_4$ and $CoFe_2O_4$ is completely different. Research shows that Co^{2+} substitution transforms the eddy current loss of the material into a magnetic resonance mode, and at the same time leads to an increase in magnetic crystal anisotropy, and thus enhancing the magnetic loss ability.

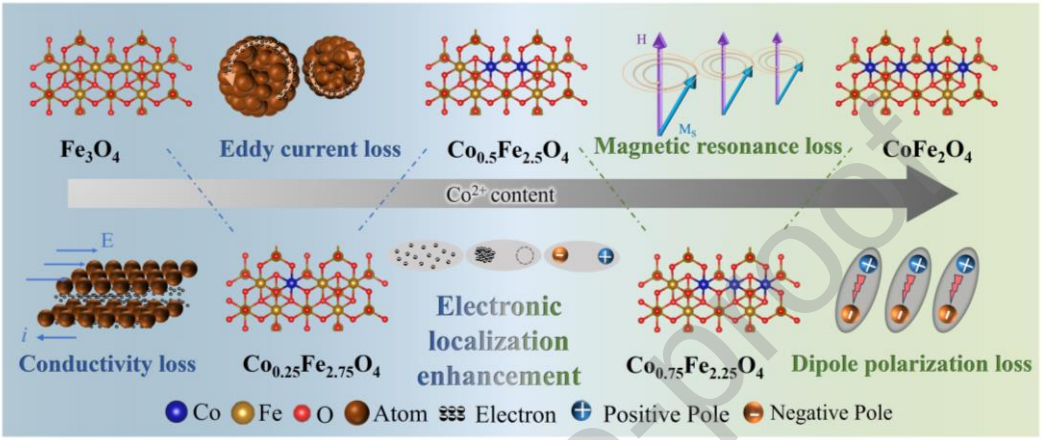


Figure 8 Schematic illustration showing the mechanisms involved during EMW absorption for the $Co_xFe_{3-x}O_4$.

4 Conclusions

In summary, a series of $Co_xFe_{3-x}O_4$ ($0 \leq x \leq 1$) absorbers were prepared using a facile hydrothermal process. The results showed that the prepared $Co_xFe_{3-x}O_4$ had a perfect hollow microsphere structure with an average size of 180-250 nm and a shell thickness of approximately 24 nm. As the value of x gradually increased from 0 to 1, the absorption performance of $Co_xFe_{3-x}O_4$ decreased and then increased. Among them, the EMW absorption performance of $CoFe_2O_4$ is significantly improved compared to Fe_3O_4, achieving a minimum reflection loss value of -72.3 dB and an effective absorption bandwidth of 3.2 GHz at 2.67 mm. This study confirms that for hollow sphere $Co_xFe_{3-x}O_4$, the changes in the electronic structure during Co^{2+} substitution can lead to transitions in the absorption mechanism, which is the underlying reason for the changes in absorption performance. This study has significance for a deeper understanding of the intrinsic relationship between the absorption performance of ferrite and its electronic structure.

Credit authorship contribution statement

Shu Li: Conceptualization, Writing - original draft, Data curation, Investigation. **Shaoli Guo:** Writing - review & editing, Methodology, Validation. **Guoxiang Chen:** Software, Supervision. **Yan Cui:** Investigation, Methodology. **Hangfan Yang:** Data curation. **Jie Qiu:** Conceptualization. **Zelin Wang:** Investigation. **Minghan Dai:** Validation. **Shuai Liu:** Resources, Funding acquisition, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Declaration of Interest Statement

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Highlights

- Monodisperse $Co_xFe_{3-x}O_4$ ($0 \leq x \leq 1$) hollow spheres were synthesized by a simple and facile hydrothermal method.
- Research shows that as Co^{2+} increases, the dielectric loss mechanism transitions from conductivity loss to polarization relaxation loss.
- As Co^{2+} increases, the magnetic loss mechanism undergoes a transition from eddy current loss to a natural resonance mode.
- $CoFe_2O_4$ possesses the best electromagnetic wave absorption performance, with the RL_{min} up to -72.3 dB at 2.67 mm.

Journal Pre-proof